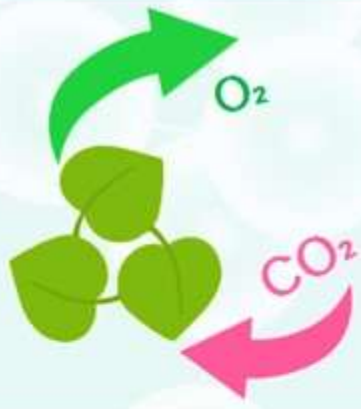


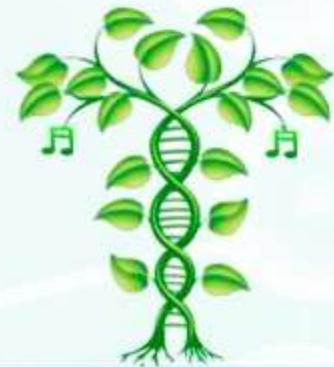


BIOLOGY



CLASSIFICATION OF ORGANISMS - 1

MODULE - 13



WHY DO WE CLASSIFY DIFFERENT FORMS OF LIFE?



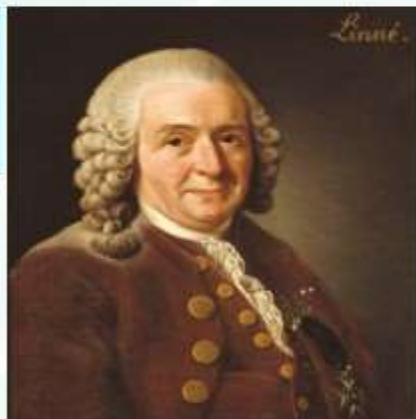
- *Each form of life has different characteristics.*
- *Helps to understand evolutionary changes between organisms*
- *Helps to identify and group newfound organisms under a proper category.*
- *Easy study of the characteristics.*



CLASSIFICATION OF ORGANISMS

TWO KINGDOM
Carl Linnaeus

- PLANT
- ANIMAL



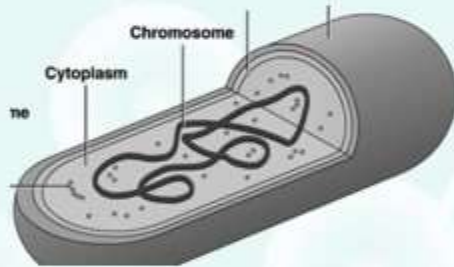
FIVE KINGDOM
R. Whittaker

- MONERA
- PROTISTA
- FUNGI
- PLANTAE
- ANIMALIA

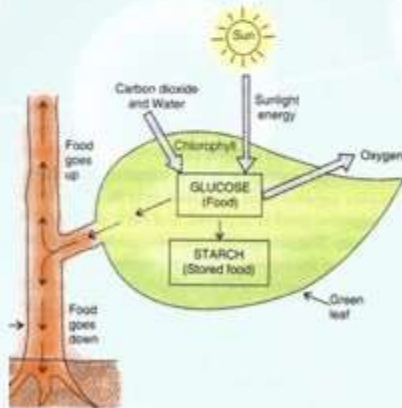


CLASSIFICATION CRITERIA

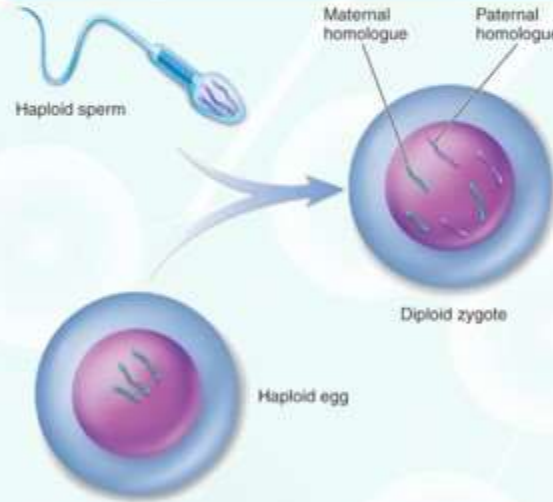
CELL STRUCTURE



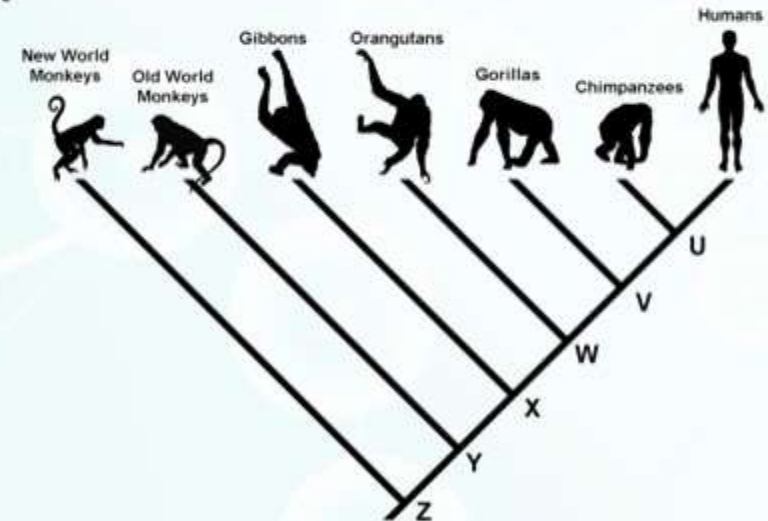
MODE OF NUTRITION

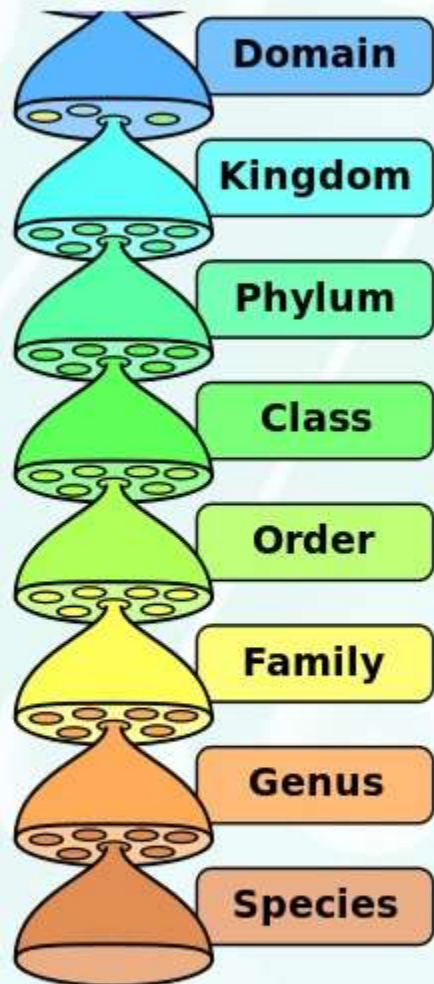


REPRODUCTIVE RELATIONSHIPS



EVOLUTIONARY RELATIONSHIPS





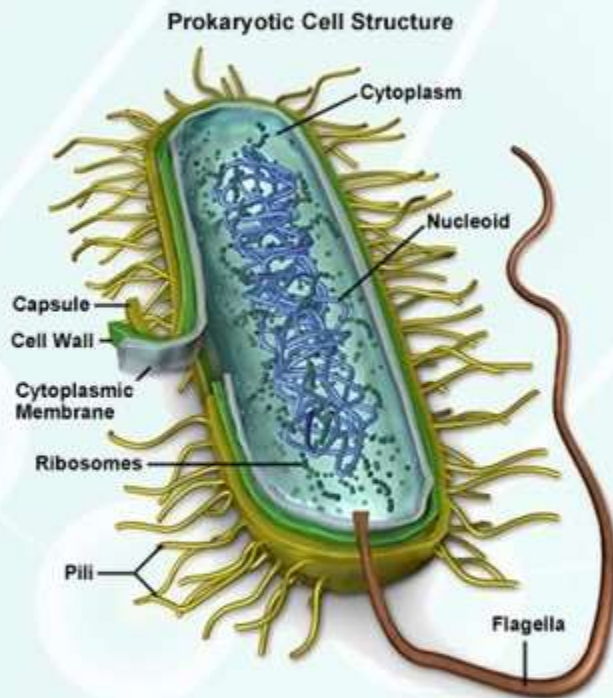
HIERARCHY OF MODERN CLASSIFICATION

- In order to understand how all living organisms are related, they are arranged into different groups.
- The more features that a group of animals share, the more specific the group is.
- Animals belong to a number of different groups, starting with the animal kingdom.
- Animals are given scientific names so that people all around the world can communicate about animals.
- This is called Binomial Nomenclature.

Mnemonic - Did King Phillip Came Over From Great Spain?



KINGDOM MONERA



- ***All kinds of bacteria***
- ***Cyanobacteria***
- ***Mycoplasma***

- They are prokaryotes.
- Cell wall is non-cellular and is composed of polysachharides and amino acids.
- Nuclear membrane is absent.
- Cellular level organisation.
- They are autotrophic and heterotrophic. Chemosynthesis and photosynthesis can be seen among autotrophs.
- Heterotrophs are saprophytic or parasitic.



KINGDOM PROTISTA



- They are eukaryotes.
- Cell wall present in some organisms.
- Nuclear membrane is present.
- Cellular level organisation.
- Autotrophic and heterotrophic.
- Autotrophs show photosynthesis

- *Amoeba*
- *Euglena*
- *Paramecium*



KINGDOM FUNGI



- They are eukaryotes.
- Cell wall is present but is not made up chitin and not of cellulose.
- Nuclear membrane is present.
- Cellular level organization.(Multicellular + loose tissue)
- Mode of nutrition is Heterotrophic which includes saprophytic and parasitic.

- *Aspergillus*
- *Penicillium*
- *Agaricus*



KINGDOM PLANTAE



- They are eukaryotes.
- Cell wall is made up of cellulose.
- Nuclear membrane is present.
- Tissue level and organ level organization.
- Mode of nutrition is Autotrophic (photosynthesis)



KINGDOM PLANTAE

THALLOPHYTA



- *Algae*
- *Spirogyra*
- *Ulothrix*
- *Cladophora*
- *Chara*

BRYOPHYTA



- *Moss*
- *Marchantia*

PTERIDOPHYTA



- *Fern*
- *Marsilea*

ANGIOSPERMS



- *All monocots and dicots*

GYMNOSPERMS

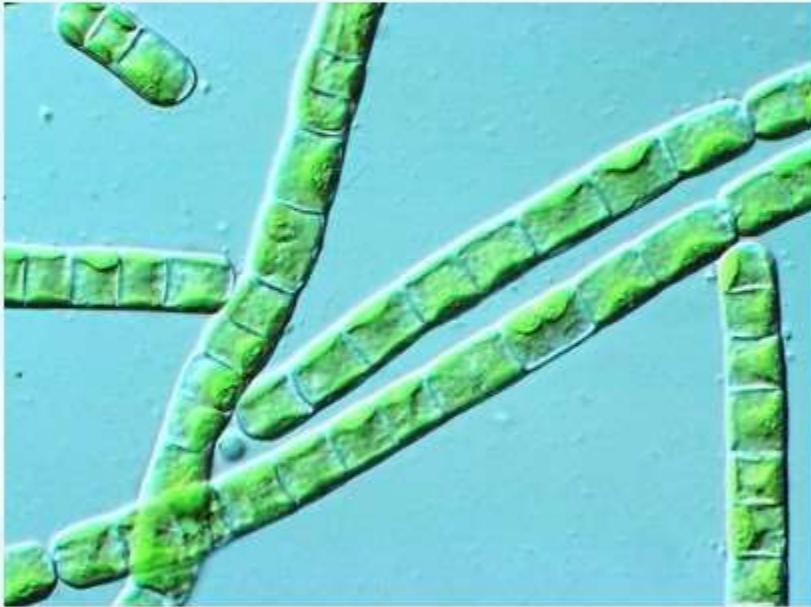


- *Pines*
- *Deodar*



KINGDOM PLANTAE

THALLOPHYTA



- Commonly algae
- No well differentiated body design(thallus).
- Predominantly aquatic.

- *Algae*
- *Spirogyra*
- *Ulothrix*
- *Cladophora*
- *Chara*



KINGDOM PLANTAE

BRYOPHYTA



- Amphibians of the plant kingdom.
- Stem and leaf life structures are differentiated
- No tissue for conduction of water.
- *Moss (Funaria)*
- *Marchantia*



KINGDOM PLANTAE

PTERIDOPHYTA



- Plant body is differentiated into roots, stems and leaves.
- Specialized tissue for conduction of water.
- Do not produce seeds.
- *Fern*
- *Marsilea*



KINGDOM PLANTAE

THALLOPHYTA



BRYOPHYTA



PTERIDOPHYTA



ANGIOSPERMS



GYMNOSPERMS



- *All monocots and dicots*

- *Pines*
- *Deodar*

- *The thallophytes, bryophytes and pteridophytes all bear naked embryos that are called spores.*
- *They have hidden reproductive organ called cryptogamiae.*



KINGDOM PLANTAE

GYMNOSPERMS



- These plants are perennial, evergreen and woody.
- They bear naked seeds
- Gymno- naked; sperms-seeds
- *Pines*
- *Deodar*



KINGDOM PLANTAE

ANGIOSPERMS



- They are all flowering plants.
- The seeds are developed inside an ovary which is modified into a fruit.
- The embryos have seed leaves or cotyledons.
- They are divided into monocot and dicot plants.



KINGDOM ANIMALIA



- They are eukaryotes.
- Cell wall is absent.
- Nuclear membrane is present.
- Tissue level, organ level and organ system level (different levels in different organisms)
- Mode of nutrition is Heterotrophic which includes holozoic, saprophytic and parasitic



KINGDOM ANIMALIA

1 PORIFERA

2 CNIDARIA

3 PLATYHELMINTHES

4 NEMATODA

5 MOLLUSCA

6 ANNELIDA

7 ARTHROPODA

8 ECHINODERMATA

9 PROTOCHORDATA

10 VERTEBRATA

Mnemonic - Prince Charles Plays New Music And Always Eats Perfect Vegetables.



WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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